THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET D

Chapters 1 & 2: **Chemical Reactions and Equations, Acids, Bases and Salts** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The silver salt used in black-and-white photography is: 1

(a) silver nitrate **(b)** silver sulphide **(c)** silver oxide **(d)** silver bromide

2. In the electrolysis of water, the gas collected at the anode is: 1

(a) hydrogen **(b)** oxygen **(c)** nitrogen **(d)** chlorine

3. A black coating forms on copper heated in air. Copper has been: 1

(a) decomposed **(b)** neutralised **(c)** reduced **(d)** oxidised

4. A student states: “Mass of products is less than that of reactants because gases escape.” The correct response is: 1

(a) Correct
(b) Correct for all reactions
(c) Incorrect — mass is conserved; it appears less only if gases escape an open system
(d) Cannot be determined

5. Which gas is evolved when dilute hydrochloric acid reacts with zinc? 1

(a) carbon dioxide **(b)** chlorine **(c)** oxygen **(d)** hydrogen

6. An aqueous solution with pH 13 is: 1

(a) strongly basic **(b)** weakly basic **(c)** neutral **(d)** strongly acidic

7. The indicator that is red in acid and yellow in base is: 1

(a) methyl orange **(b)** red litmus **(c)** turmeric **(d)** phenolphthalein

8. Bees inject which acid when they sting? 1

(a) acetic acid **(b)** citric acid **(c)** oxalic acid **(d)** methanoic acid

9. The salt NH_4Cl is formed from: 1

(a) NaOH and HCl **(b)** NH_4OH and H_2SO_4
(c) KOH and HCl **(d)** NH_4OH and HCl

10. A student states: “Bases produce OH^- ions in water.” The correct response is: 1

(a) Correct **(b)** Incorrect
(c) Only in acids **(d)** Cannot be determined

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Write two observations when lead nitrate solution is added to potassium iodide solution. 2

12. Why does aluminium not corrode readily even though it is a reactive metal? 2

13. What is a family of salts? Give two examples. 2

14. Why is a dilute acid a weaker acid than a concentrated acid of the same kind? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. List three observations when dilute hydrochloric acid is added to zinc granules, and write the equation. 3

16. What is the difference between hydrated and anhydrous salts? Give one example of each. 3

17. What is the role of sunlight in the decomposition of silver bromide? Write the equation and one use of this reaction. 3

OR

Balance and state the type: 3

(i) $Na_2CO_3 + HCl \longrightarrow NaCl + H_2O + CO_2$

(ii) $KClO_3 \longrightarrow KCl + O_2$

(iii) $Fe + Cl_2 \longrightarrow FeCl_3$

18. Why is pH important for our digestive system? What happens during indigestion and how is it treated? 3

OR

What happens when 3

(i) dilute HCl is added to iron filings

(ii) sodium carbonate solution is added to dilute HCl

(iii) NaOH solution is added to HCl with phenolphthalein?

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Why is it necessary to balance chemical equations? 5

(ii) Balance:

(a) $Fe + H_2O \longrightarrow Fe_3O_4 + H_2$

(b) $C_3H_8 + O_2 \longrightarrow CO_2 + H_2O$

(c) $NH_3 + O_2 \longrightarrow NO + H_2O$

OR

Describe the electrolysis of water: 5

(i) the set-up

(ii) the gases formed and how to test them

(iii) their volume ratio and the reason

(iv) the type of reaction.

20. (i) What is the pH of 5

(a) gastric juice

(b) pure water

(c) milk of magnesia — approximately?

(ii) What is the importance of pH in the survival of animals?

(iii) Why does acid rain damage plants?

OR

(i) What are strong and weak bases? Give one example each. 5

(ii) What happens when

(a) NaOH is dissolved in water

(b) an alkali is diluted?

(iii) Why should we not add water to concentrated acid?

END OF QUESTION PAPER Total: 40 Marks

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